

GRANT SWIFT

Mechanical Engineering Student · University of Oklahoma

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EDUCATION

University of Oklahoma

B.S. Mechanical Engineering

Norman, OK

Expected May 2028 · GPA: 3.6 / 4.0

Relevant coursework: Statics, Dynamics, Thermodynamics, Design & Manufacturing Processes

EXPERIENCE

Manufacturing Engineering Intern

Summers 2025 & 2026

SAGA Manufacturing

- Inspected 100+ machined aluminum components daily against client drawings toleranced to 0.005 in for aerospace and defense contracts (Lockheed Martin)
- Resolved a tolerance stack-up failure where in-spec brackets would not assemble; traced it to interface tolerances and corrected the process so parts held the print and fit the assembly
- Ran CNC and manual machining to aerospace compliance specs, reading GD&T-toleranced blueprints
- Caught drifting dimensions early with in-process caliper inspection, preventing full-batch scrap

PROJECTS

Stockwell

Solo build

Full-Stack Retail Operations Platform

- Designed and shipped a multi-tenant SaaS app automating staff scheduling and inventory restocking for small retailers: ~29,000 lines across 156 files
- Built an AI scheduling engine with an independent verification layer that repairs double-bookings and untrained assignments before publishing
- Integrated Lightspeed, Square, and Stripe via OAuth 2.0; enforced per-business data isolation validated by 300+ automated tests
- Stack:** JavaScript, Node.js, Express, React, SQL, REST APIs, Docker

Sooner Racing (Formula SAE)

Team

Cooling System Integration · Powertrain

- Sized and fit a temporary radiator so the team could run the car in on-track testing before the final cooling package was finished
- Designed and fabricated the mounting brackets and support frame to package the radiator in the chassis and keep it secure under track loads
- Kept the engine within safe operating temperature through test sessions, keeping the team's testing schedule on track

3D-Printed Paddle Shifters

Personal

SolidWorks Design & Fabrication

- Reverse-engineered and modeled extended paddles to fit factory Audi A7 mounts with no OEM reference
- Iterated through 6 revisions using tolerance test coupons before committing material; tuned print orientation and infill for strength
- Parts survive 130°F+ cabin heat without warping; wet-sanded and clear-coated to match factory trim

Precision Woodturning

Ongoing

Handcrafted Pens

- Produce finished pens from raw blanks on the lathe: drill, epoxy brass reference tubes, turn to diameter, finish, and press-fit to hardware
- Hold a repeatable process across 1000+ pieces, keeping consistent diameters and press-fit tolerances so every unit assembles to spec

LEADERSHIP & ACTIVITIES

LaGrone Academy

Lead Design Engineer

- Led design on an RC car build project and managed budget and system integration for an automated trash-pickup project

TECHNICAL SKILLS

CAD & Design: SolidWorks, AutoCAD

Manufacturing: CNC machining, GD&T & blueprint reading, 3D printing, inspection & measurement

Software: JavaScript, Node.js, React, SQL, Git

Analysis: Statics, Dynamics, Thermodynamics

CERTIFICATIONS

AWS Cloud Practitioner Essentials (2026) · OSHA (2024) · CAD Certified (2024)